Module-1

1. Environment means
   - **a. Surrounding (Biotic + abiotic), in which organisms live**
   - b. Atmosphere around one self
   - c. Sum total of social, economical, biological behavior of animals
   - d. Sum total of developmental activities around
2. Unsustainable development is a result of
   - a. Importance given to only economic and social patterns
   - b. No concern was given to maintain quality and quantity of the resources
   - c. Orientation only towards the importance of the economy of the country
   - **d. All the above**
3. Which of the following is not the meaning of ecosystem
   - a. Unit where in all organism live a healthy life
   - **b. A small unit that can be self sufficient**
   - c. Co-existence of diverse things by mutual adjustment
   - d. A unit which includes all of the organisms in a given area interacting with the physical environment to form a natural unit of stability
4. Which of it is not an example for an ecosystem
   a. Forest b. Desert **c. Water** d. Grassland
5. The factors responsible for stable ecosystem are – balance between
   - a. Predators and prey
   - b. Vegetation, herbivores and carnivores
   - c. Competing species and biotic factors
   - **d. All the above**
6. Tertiary consumers are
   a. Predators **b. Predators of predator's** c. Prey d. Herbivores
7. Nutrient cycling is most related to appropriately
   - a. Energy, waste, nutrients
   - **b. Autotrophs, nutrients, decomposers**
   - c. Light, weight, nutrients
   - d. None of these
8. Light – a form energy in an ecosystem according to 1st law of energy consumption cannot be
   - a. Transformed into work
   - b. Transformed into heat
   - **c. Destroyed completely**
   - d. Transformed into potential energy of food
9. Which of the two are true
   1. Energy is lost as heat in an energy flow of an ecosystem
   2. Nutrients are lost as energy is a nutrient cycle
   3. Nutrients are recycled in a nutrient cycle of an ecosystem
   4. Energy is recycled in an energy flow of an ecosystem

   a. 2 & 4 **b. 1 & 3** c. 1 & 2 d. 3 & 4

10. Habitat refers to
    a. Physical conditions of the place where organism live
    b. Chemical conditions of the place where organism live
    c. Both (a) and (b)
    d. Neither (a) nor (b)
11. Important physical factor responsible for a good habitat of a organisms
    a. Light c. Humidity
    b. Temperature **d. All the above**
12. Which of the following is not a prominent chemical responsible for a good habitat
    a. Oxygen **c. SO_2**
    b. Carbondioxide d. Nutrients (Minerals)
13. The term 'Environment' has been derived from the French word ---------- which means to encircle or surround
    a. **Environ** c. Geo
    b. Oikos d. Aqua
14. The objective of environmental education is
    a. Raise consciousness about environmental conditions
    b. To teach environmental appropriate behavior
    c. Create an environmental ethic
    d. All the above
15. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
    a. **Atmosphere** c. Hydrosphere
    b. Lithosphere d. Biosphere
16. Which of the following components of the environment are effective transporters of matter?
    a. **Atmosphere and hydrosphere** c. Hydrosphere and Lithosphere
    b. Atmosphere and lithosphere d. Biosphere and lithosphere
17. Biosphere is
    a. The solid shell of inorganic materials on the surface of the earth
    b. The thin shell of organic matter on the surface of earth comprising of all the living thing
    c. The sphere which occupies the maximum volume of all of the spheres
    d. All of the above
18. Atmosphere consists of 79% of Nitrogen and 21% of Oxygen by
    a. **Volume** c. Density
    b. Weight d. All the above
19. Which of the following is a biotic component of an ecosystem?
    a. **Fungi** c. Temperature
    b. Solar light d. Humidity
20. In an ecosystem, the flow of energy is
    a. Bidirectional c. Cyclic
    b. **Unidirectional** d. Multidirectional
21. The organisms who directly feed on producers are called
    a. **Herbivores** c. Decomposers
    b. Carnivores d. Saprophytes

22. The sequence of eating and being eaten in an ecosystem is called
    a. **Food chain** c. Hydrological cycle
    b. Carbon cycle d. Anthroposystem
23. Which of the following is a producers in an ecosystem
    a. Plants and some bacteria capable of producing their own food
    b. Animals c. Human beings d. Fish
24. The driving force for the circulation of water in the hydrological cycle is provided by _____
    a. Ozone **b. Sun** c. Moon d. Mercury
25. Which of the following statements is false?
    a. Inorganic nutrients are recycled in an ecosystem
    b. Energy flows through the ecosystem in the form of carbon-carbon bonds
    c. Energy is recycled in an ecosystem
    d. Respiration process releases energy
26. The largest reservoir of nitrogen in our planet is
    a. Oceans c. Biosphere
    b. **Atmosphere** d. Fossil fuels
27. In aquatic ecosystem phytoplankton can be considered as a
    a. Consumer c. Saprotrophic organisms
    b. **Producer** d. Macroconsumer
28. The Greek word "Oikos" means ----------
    a. surroundings **b. Habitation** c. Food chain d. Food web
29. The basic requirements of human beings are provided by
    a. Industrialization **c. Nature**
    b. Agriculture d. Urbanization
30. Environment is the life support system that includes
    a. Air c. Land
    b. Water **d. All of the above**
31. Organisms which feed directly or indirectly on producers are called
    a. Prey c. Decomposers
    b. **Consumers** d. Detritus
32. The primary producers in a forest ecosystem are
    a. Chlorophyll containing trees and plants
    b. Herbivores c. Carnivores d. Bacteria and other microorganisms
33. Abiotic component includes
    a. Soil c. Water
    b. Temperature **d. All of the above**
34. Which of the following statement is true
    a. Green plants are self nourishing
    b. Producers depends on consumers
    c. Biotic components includes all non-living components
    d. Herbivores depend on carnivores
35. The motive forces responsible for the circulation of water in hydrological cycle are _____
    a. Thermal energy b. Earths gravitation **c. Both a and b** d. Neither a nor b

36. Primary consumer is
    a. **Herbivores** b. Carnivores c. Macro consumers d. Omnivores
37. A predator is
    a. An animal that is fed upon
    b. An animal that feeds upon another animal
    c. Animal that feeds upon both plants and animals
    d. A primary consumer
38. The measure of the relative abundance of the different species making up the richness of an area is called _____
    a. Species evenness b. Species richness c. Keystone species d. None of these
39. The word 'Environment' derived from
    a. Greek **b. French** c. Spanish d. English
40. The major atmospheric gas layer in stratosphere is
    a. Hydrogen b. Carbon dioxide **c. Ozone** d. Helium
41. Which atmospheric sphere is closest to the earth surface?
    a. **Troposphere** b. Stratosphere c. Mesosphere d. Exosphere
42. Which of the following is the terrestrial ecosystem?
    a. Forest b. Grassland c. Desert **d. All of the above**
43. Which layer acts as a protective shield for life and earth from injurious effects of the sun's ultraviolet rays
    a. Ozone layer b. Mantle c. Crust d. Core
44. Ecological pyramids are studies of
    a. Pyramid of numbers c. Pyramid of energy
    b. Pyramid of biomass **d. All of the above**
45. The availability of fresh water for direct use is about _____
    a. Less than 1% b. 3% c. 97% c. 5%
46. World Environment day is on
    a. 5th May **b. 5th June** c. 18th July d. 16th August
47. A food web consists of
    a. A portion of a food chain
    b. An organisms position in a food chain
    c. Interlocking food chains
    d. A set of similar consumers
48. Which of the following statements are true?
    a. Man is not dependent on nature
    b. Resources are unlimited, so one can use them as per one's wish.
    c. Energy can be converted from one form to another, but some percentage is lost into the environment
    d. Matter can be generated afresh. It need not be recycled or reused.
49. Hydrological cycle mainly involves
    a. **Sun and water** b. Air and water c. Animals and water d. Mountains and water
50. Which region of the atmosphere is also called as ionosphere
    a. Troposphere **b. Thermosphere** c. Exosphere d. Stratosphere
51. Hydrological cycle is related to
    a. Water and electricity **b. water cycle and balance** c. water characterization d. Hydropower

52. Which of the following is not a part of the hydrological cycle?
    a. Precipitation b. Infiltration c. Transpiration **d. Perspiration**
53. In which region of the atmosphere we can find climatic and weather changes.
    a. Stratosphere **b. Troposphere** c. Ionosphere d. Mesosphere
54. Earth is made of three distinct layers
    a. Troposphere, stratosphere, mesosphere
    b. Core, mantle, crust
    c. Ozonosphere, thermosphere, ionosphere
    d. Exosphere, heterosphere, hydrosphere
55. ---------- is termed as the life zone of the earth
    a. Atmosphere b. Hydrosphere **c. Biosphere** d. Stratosphere
56. The term Ecology was derived from _____ word
    a. French **b. Greek** c. Spanish d. Latin
57. Mesosphere is characterized by
    a. Low atmospheric pressure and cold temperature
    b. Presence of mainly nitrogen and oxygen
    c. Abrupt changes in velocity of seismic waves
    d. Tremendous temperature of around 5000^0C
58. The radius of the earth is about
    a. 6371km b. 5200km c. 3500km d. 40000km
59. Which region of the atmosphere enables us to have wireless communication?
    a. Mesosphere b. Stratosphere c. Troposphere **d. Ionosphere**
60. The definition of Environment was given by
    a. Eugine P. Odum b. Darwin c. Galilio d. Sundar Lal Bhahugun
61. Earth's day is celebrating on
    a. June 5th b. November 23rd **c. April 22nd** d. January 26th
62. Which pyramid is always upright
    a. **Energy** b. Biomass c. Number d. Food chain
63. In an ecosystem biological cycling of materials is maintained by
    a. **Producer** b. Consumer c. Decomposer d. All of these
64. The term environment was introduced by
    a. Jacob Van Verkul b. Tyler Miller c. Gilbertson **d. Chales Darwin**
65. An ecosystem is a region in which
    a. Dead organism interact with their environment
    b. Living organism do not interact with their environment
    c. **Living organisms interact with their environment**
    d. All of these
66. Many jet aircrafts fly in the stratosphere because,
    a. It is drastically changing **b. It is very stable** c. it is more turbulent d. None of these
67. The term ecosystem was proposed by
    a. Jacob Van Verkul **b. A.G. Transley** c. Costanza d. Marie Gibbs
68. The two major components of ecosystem are
    a. Adiabatic and isotropic b. Ecological and climatologic c. Cyclic and biologic **d. Abiotic and biotic**

69. Biotic components include
    a. **All living organisms** b. Water, Mineral and gasses c. self-nourishing green plants d. light, temperature
70. Guttenberg-Weichert discontinuity separates ---------- and ----------
    a. Crust, Mantle **b. Mantle, Core** c. Outer core, Inner core d. Sial, Sima layer
71. Food chain is divided into---------- basic categories
    a. Four **b. Three** c. Five d. Seven
72. The upper layer of the troposphere which gradually merges with the stratosphere is called -------
    a. Stratopause **b. Tropopause** c. Mesopause d. Mesosphere
73. Aquarium is an example of
    a. **Man-engineered ecosystem** b. Natural ecosystem c. Desert ecosystem d. Forest ecosystem
74. Which of the following is the most abundant gas in the atmosphere is
    a. Oxygen **b. Nitrogen** c. Argon d. Carbon-di-oxide
75. The composition of core is _____
    a. Oxygen and Nitrogen b. Copper and iron **c. Nickle and iron** d. Nickle and copper
76. Life on earth is supported by
    a. Atmosphere b. Solar energy c. Earth's magnetic field **d. All of these**
77. The shorter the food chain
    a. **Greater is the available food energy** c. No food energy is available
    b. Minimum is the available food energy d. None of these
78. Which of the following is a true representation of a food chain?
    a. Grass ⟶ Tiger ⟶ Rabbit ⟶ Grasshopper
    b. Tiger ⟶ Rabbit ⟶ Grasshopper ⟶ Grass
    c. **Grass ⟶ Grasshopper ⟶ Rabbit ⟶ Tiger**
    d. Rabbit ⟶ Tiger ⟶ Grass ⟶ Grasshopper
79. _____ is defined as an ecological state of a species being unique to a specific geographic location.
    a. Exotic species **b. Endemic species** c. Ecosystem d. None of these
80. _____discontinuity separates a crust composed of low density silicates rich in aluminium and potassium, from mantle composed of silicates of higher density which contain more magnesium and iron.
    a. Conrad b. Guttenberg **c. Mohorovicic** d. Ingleman
81. _____ is the forest cover to be maintained as per the National Forest Policy (1988).
    a. **67% for hills & 33% for plains** b. 37% for hills & 11% for plains
    c. 17% for hills & 23% for plains d. None of these
82. _____ is defined as the number of species represented in a specific region, landscape or an ecological community.
    a. Coevolution b. Commensalism **c. Species richness** d. Population density
83. Which of the following animals is now extinct?
    a. **Tasmanian tiger** b. Tasmanian devil c. Pademelon d. Quoll
84. Which of the following is correct
    a. **Pyramid of energy is always upright** c. Pyramid of number is always upright
    b. Pyramid of biomass is even upright d. Pyramid of energy is inverted also
85. The term Ecology was derived from Greek word ----------
    a**. Oikos** b. Environ c. Eco d. Geo

86. Seismic waves characterize the internal structure of the earth and nature of its layers, which suggest that the outer core is _____ whereas the inner core is_____
    a. Liquid, Solid b. Plastic, Liquid c. Solid, Liquid d. Liquid, Plastic
87. _____ are the 'recyclers of the biosphere'.
    a. Producers b. Consumers **c. Decomposers** d. All of these
88. The outer solid portion of the earth existing above the asthenosphere (partially molten plastic layer) is called___
    a. Lithosphere b. Mesosphere c. stratosphere d. Exosphere
89. The regions which are known for their high species richness and endemism is called as_____
    a. Restricted ecosystem **b. Biodiversity hotspots** c. Biodiversity scarcity d. Biome
90. Which layer of the atmosphere is called as coldest region in the atmosphere
    a. Troposphere b. Stratosphere **c. Mesosphere** d. Thermosphere
91. 10% law of energy transfer states that during transfer of energy from one trophic level to another in a food chain _____ % of the energy is lost to the surroundings and only _____ % is transferred to next trophic level.
    a. 90, 10 b. 10, 90 c. 50, 50 d. 25, 75
92. The variety of plant and animal species in a particular habitat is known as _____
    a. Biodiversity b. Biome c. Ecosystem d. Ecology
93. _____ is one of the most prevalent hotspots of biodiversity in India
    a. Himalayas **b. Western Ghats** c. Ganges d. None of the above
94. Which type of food chain starts from dead matter to micro-organisms?
    a. Predator food chain b. Parasitic food chain **c. Saprophytic food chain** d. Grazing food chain
95. _____ is not generally seen in biodiversity hotspots.
    a. Endemism b. Species richness c. Loss of diversity **d. Lesser interspecific competition**
96. In the context of conservation WWF stands for
    a. World Wildlife Federation b. World Wildlife Forum **c. World Wildlife Fund** d. none of these
97. The maximum number of National Parks are located in
    a. Karnataka **b. Madhya Pradesh** c. Tamil Nadu d. Maharashtra
98. Who coined the word Biodiversity hotspot in 1988?
    a. Charles Darwin b. Saleem Ali **c. Norman Myers** d. None of these
99. Silent Valley having rare plants and animals is located in
    a. Uttaranchal **b. Kerala** c. Karnataka d. Jammu and Kashmir
100. A food web consist of
    a. Single food chain c. **More than one interconnected food chain**
    b. Two separate linear food chain d. None of these